Ali Darwich Ismail

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SUMMARY

An aspiring robotics engineer with hands-on experience in robotics design, programming, embedded systems, and control systems. Two-time international robotics competition winner and experienced robotics instructor. Seeking professional and hands-on experience to contribute to robotics projects in automation, biomechatronics, control, and embedded systems.

EDUCATION

Bachelor of Engineering in Mechatronics Engineering *Sep 2023–Present*
Lebanese American University, Byblos, Lebanon *Expected Graduation: May 2028*
Lebanese Baccalaureate of General Science *Sep 2020–Jun 2023*
Rafic Hariri High School, Saida, Lebanon
Graduated with Distinction (16/20)

PROFESSIONAL EXPERIENCE

Robotics Intern at Camerabotics, Beirut, Lebanon *February 2026 – August 2026*

- Designed a UART communication system between a daisy chain of custom STM32 PCBs and a Raspberry Pi.
- Debugged and extended C++ motor control firmware on STM32, implementing position/velocity control, low-level UART protocol, and a flash saving memory map.
- Built a Python motor control API and the high-level UART protocol and deployed them on a Raspberry Pi.

LAU Robotics Summer Camp, Lead Organizer & Robotics Instructor *July 2025, July 2024*

- Designed, delivered and monitored robotics camp sessions for 40+ students using Lego Spike Prime.
- Coordinated camp logistics and managed a team of 4 instructors and 4 volunteers.

PROJECTS

FrameLock: Cinematographic robot with real-time face detection and tracking; Controller and Raspberry Pi Embedded system Designer, Face detection and tracking developer *November 2025*

- Applied face detection algorithm using OpenCV on a Raspberry Pi and developed the focal distance algorithm to get the distance between the robot and the face.
- Designed and tuned dual PID controllers to stabilize robot orientation and maintain smooth cinematic tracking with switchable subjects.

2-DoF Biomimetic Robotic Fish; Project Leader, Embedded System Developer *May 2025*

- Led the team and coordinated the project workflow.
- Designed and developed the reverse kinematics and encoder algorithms and applied them to a cascade PID motor controller.

Control+Farm: An Automatic farming Robot; Project Leader, MyRio and Arduino Embedded System Designer *August 2025*

- Designed and programmed an autonomous robot for seeding and watering crops, using IMU data for stability monitoring and fall detection.
- Implemented automated irrigation scheduling, reducing manual labor for small-scale farming.

TECHNICAL SKILLS

- **Programming**: C/C++, Java, MATLAB/Simulink, Python, Linux, ROS2, Git and GitHub.
- **Robotics/Embedded Systems**: Sensors and Sensor Fusion, Control Systems, Arduino, MyRio, Raspberry Pi, STM32, open CV.
- **Software Tools**: AutoCAD, SolidWorks, Microsoft Office, LabView, MATLAB, VSCode.

AWARDS & ACHIEVEMENTS & CERTIFICATIONS

- *2x Middle East Champion, Do Your Bit Competition by BBC (2021, 2022).* *July 2021, July 2022*
- *Best Programmer, Do Your Bit National Competition (2023).* *July 2023*
- *Programming in C++: A Hands-on Introduction specialization, Coursera*
- *Arduino For Beginners course offered by Packt, Coursera*
- *Introduction to Programming with MATLAB, Coursera*
- *Robotics specialization offered by the University of Pennsylvania, Coursera*

LANGUAGES

- Arabic (Native)
- English (C2)
- French (B2)